

Data Dictionary

Mortgage Delinquency Risk Analysis — Fannie Mae Single-Family Loan Performance Dataset (2020 Q1–Q2)

This dictionary documents every field carried forward from the raw Fannie Mae source into this project's analytical tables, plus every feature engineered on top of them. Raw column positions are mapped against the business field names used throughout the SQL pipeline and dashboard. Full raw-to-business mapping is also available in machine-readable form in [Column_Mapping.xlsx](#).

Raw Source Fields

Selected from the 108-column raw Fannie Mae extract during the mortgage_analysis_q1 / mortgage_analysis_q2 table build. These 30 fields carry the loan attributes required for credit risk analysis.

Raw Column	Field	Type	Description
column2	loan_id	TEXT	Unique loan identifier
column3	monthly_reporting_period	TEXT	Reporting period for this loan-month observation
column24	credit_score	INTEGER	Borrower credit score at origination
column23	dti_ratio	NUMERIC	Debt-to-income ratio at origination
column22	num_borrowers	INTEGER	Number of borrowers on the loan
column26	first_time_homebuyer	TEXT	First-time homebuyer indicator (Y/N)
column20	orig_ltv	NUMERIC	Original loan-to-value ratio
column21	orig_cltv	NUMERIC	Original combined loan-to-value ratio
column8	orig_interest_rate	NUMERIC	Original interest rate on the loan
column13	orig_loan_term	INTEGER	Original loan term in months
column27	loan_purpose	TEXT	Purpose of loan: Purchase, Cash-Out Refi, No Cash-Out Refi
column35	amortization_type	TEXT	Amortization type (fixed rate)
column37	interest_only_flag	TEXT	Interest-only loan indicator (Y/N)
column4	origination_channel	TEXT	Channel through which loan was originated
column28	property_type	TEXT	Type of property securing the loan
column29	num_units	INTEGER	Number of units in the property
column30	occupancy_status	TEXT	Owner-occupied, second home, or investment
column31	property_state	TEXT	US state where property is located
column32	msa_code	TEXT	Metropolitan Statistical Area code
column33	zip_short	TEXT	First three digits of property ZIP code
column86	property_valuation_method	TEXT	Method used to value property at origination
column10	orig_upb	NUMERIC	Original unpaid principal balance
column12	current_upb	NUMERIC	Current unpaid principal balance for this period

Raw Column	Field	Type	Description
column16	loan_age	INTEGER	Age of loan in months as of reporting period
column14	origination_date	TEXT	Month/year loan was originated
column40	delinquency_status	TEXT	Delinquency status code for this reporting period
column41	payment_history	TEXT	Encoded payment history string
column42	modification_flag	TEXT	Indicates whether loan has been modified (Y/N)
column44	zero_balance_code	TEXT	Reason code if loan balance reduced to zero
column46	upb_at_removal	NUMERIC	Unpaid principal balance at time of loan removal
column52	foreclosure_date	TEXT	Date foreclosure was completed, if applicable

Engineered Features

Derived within mortgage_typed_fe_v to support credit risk segmentation and the delinquency flags used throughout the dashboard and reporting layer.

Field	Type	Logic
credit_score_band	TEXT	Very High Risk: 620–639 High Risk: 640–679 Moderate Risk: 680–719 Low Risk: 720–759 Very Low Risk: 760+
ltv_band	TEXT	Very Low Risk: <60 Low Risk: 60–79 Moderate Risk: 80–89 High Risk: 90+ (PMI threshold at 80%)
dti_band	TEXT	Low Risk: ≤28 Moderate Risk: 29–36 High Risk: 37–43 Very High Risk: 44+ (28/36 rule, 43% Fannie Mae max)
risk_factor_count	INTEGER	Sum of three flags: credit_score < 680 (+1), orig_ltv > 90 (+1), dti_ratio > 43 (+1). Range 0–3.
risk_segment	TEXT	Low Risk: risk_factor_count = 0 Moderate Risk: risk_factor_count = 1 High Risk: risk_factor_count ≥ 2
delinquency_flag	INTEGER	1 if delinquency_status is not '00' (current) and not 'XX' (unknown), else 0
serious_delinquency_flag	INTEGER	1 if delinquency_status is 90+ days past due (status codes 03–25), else 0. Aligned to Basel/IFRS 9 default threshold.

Full SQL logic: 03_SQL/mortgage_risk_analysis.sql | Machine-readable mapping: 04_Data/Column_Mapping.xlsx | Source and license notes: 04_Data/Dataset_Source.md